Speech Recognition (DSAI 456)

Lecture 6: HMMs and GMMs by Hand

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Lecture 6, Monday 2 November 2026.

Agenda

Objectives: by the end of this lecture you should be able to fill a forward and a Viterbi trellis by hand for an HMM you have not seen before, score a frame with a Gaussian and with a mixture, run one EM step by hand, and explain how a GMM-HMM is trained.

- **Part 1, HMMs by hand**: forward, Viterbi and backtrace on new examples
- **Part 2, GMMs by hand**: human k-means, Gaussian scores, shares, EM, and training the GMM-HMM
- **Part 3, Putting it together**: decoding, the limits of the GMM, and an exam-style problem

Important
Most of today is practice: every **Practice** slide is done in pairs, on paper, before the answer appears. Keep a calculator ready.

Recap: Forward and Viterbi Side by Side

	Forward (sum)	**Viterbi** (max)
A cell holds	$\alpha_t(j)$: **sum** over all paths that end in j at time t	$\delta_t(j)$: probability of the **best** such path
Start	$\alpha_1(j) = \pi_j \, b_j(x_1)$	$\delta_1(j) = \pi_j \, b_j(x_1)$
Step	$\alpha_t(j) = \left[\sum_i \alpha_{t-1}(i) \, a_{ij} \right] b_j(x_t)$	$\delta_t(j) = \left[\max_i \delta_{t-1}(i) \, a_{ij} \right] b_j(x_t)$, and store $\psi_t(j)$
End	$P(X \mid \lambda) = \alpha_T(N)$	$P^* = \delta_T(N)$, then backtrace
Answers	how likely is this word?	which frames belong to which state?

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Answers	how likely is this word?	which frames belong to which state?

One cell, four moves: (1) find the arrows that come **into** the cell, (2) multiply each predecessor by its arrow, (3) **add** them (forward) or **keep the largest** (Viterbi), (4) multiply by the emission of the **current** frame.

Part 1

HMMs by Hand

Forward, Viterbi and backtrace on new examples

Practice 1: Score One Path (3 min)

Same *salaam* model as Lectures 4 and 5: $\pi = (1, 0)$; $a_{ss} = 0.5$, $a_{sa} = 0.5$, $a_{aa} = 1$; /s/ emits h 0.8, v 0.2; /a/ emits h 0.2, v 0.8.
New sounds: $X = (h, h, v, v)$.

Compute $P(X, Q \mid \lambda)$ for each state sequence Q:

a. s s a a **b.** s a a a **c.** s a s a **d.** a a a a

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Compute $P(X, Q \mid \lambda)$ for each state sequence Q:

a. s s a a **b.** s a a a **c.** s a s a **d.** a a a a

Q	start × emit, then (move × emit) for each frame	$P(X, Q)$
s s a a	$1 \cdot 0.8 \cdot 0.5 \cdot 0.8 \cdot 0.5 \cdot 0.8 \cdot 1 \cdot 0.8$	**0.1024**
s a a a	$1 \cdot 0.8 \cdot 0.5 \cdot 0.2 \cdot 1 \cdot 0.8 \cdot 1 \cdot 0.8$	0.0512
s a s a	the move /a/ → /s/ has probability 0	0
a a a a	$\pi_a = 0$: a word cannot start in /a/	0

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A path is **one long product**. A single zero anywhere (in π, A or B) kills the whole path.

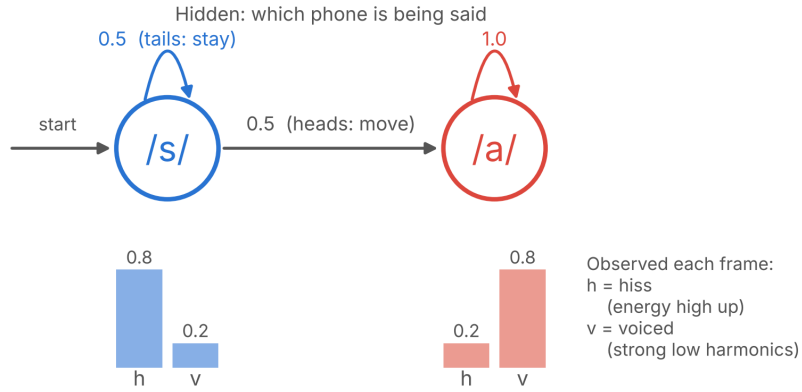
Practice 2: Forward and Viterbi Together (8 min)

Same model, $X = (h, h, v, v)$. Fill **two** numbers per cell: α (add) and δ (keep the larger). Then backtrace.

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- $t = 1$ (h): $\alpha_1 = \delta_1 = (0.8,\ 0)$

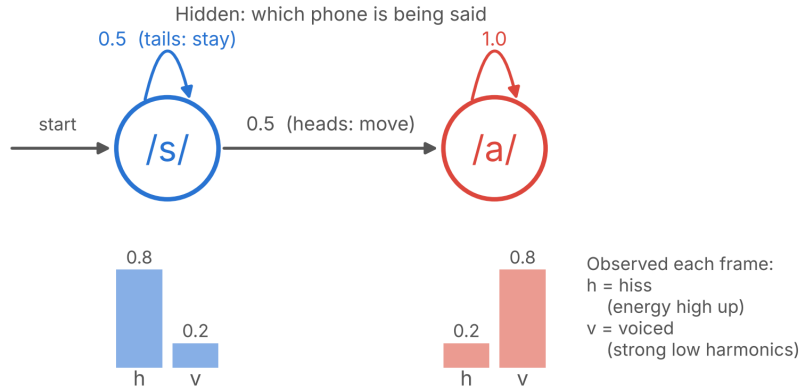


The model: transitions A on the arrows, emissions B in the bars under each state.

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- $t = 1$ (h): $\alpha_1 = \delta_1 = (0.8,\ 0)$
- $t = 2$ (h): $\alpha_2(s) = 0.8 \cdot 0.5 \cdot 0.8 = 0.32$; $\alpha_2(a) = \left[\alpha_1(s)\, a_{sa} + \alpha_1(a)\, a_{aa}\right] b_a(h) = (0.8 \cdot 0.5 + 0 \cdot 1) \cdot 0.2 = 0.08$. Only one real way in, so $\delta_2 = \alpha_2$.

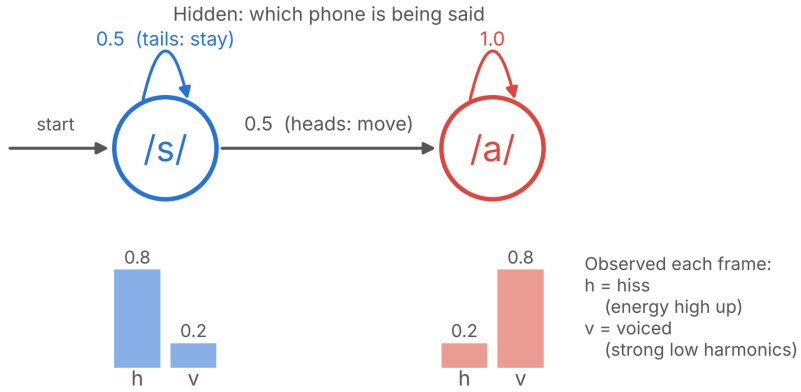


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- $t = 3$ (v): $\alpha_3(s) = 0.032$; $\alpha_3(a) = (0.32 \cdot 0.5 + 0.08 \cdot 1) \cdot 0.8 = 0.192$; $\delta_3(a) = \max(0.16,\ 0.08) \cdot 0.8 = 0.128$, from /s/

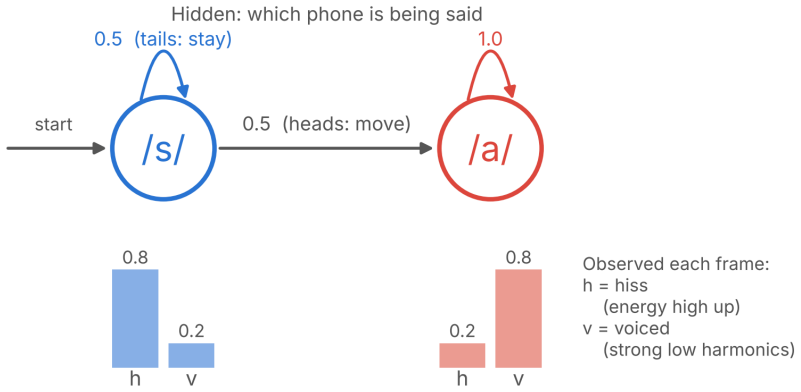


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- $t = 4$ (v): $\alpha_4(a) = (0.032 \cdot 0.5 + 0.192 \cdot 1) \cdot 0.8 = \mathbf{0.1664}$; $\delta_4(a) = \max(0.016,\ 0.128) \cdot 0.8 = \mathbf{0.1024}$, from /a/

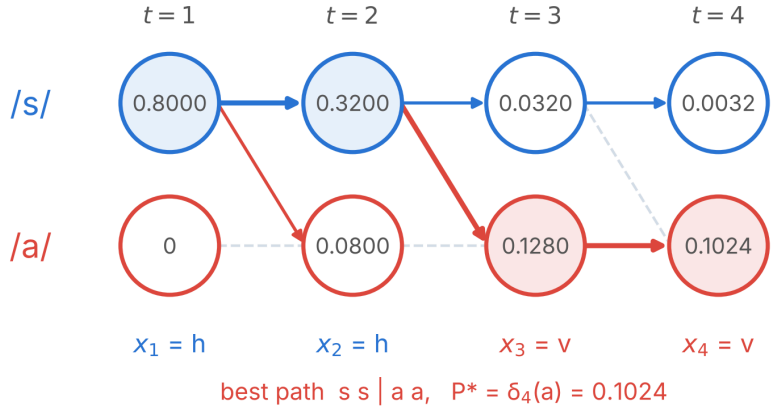


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Practice 2: Forward and Viterbi Together (8 min)

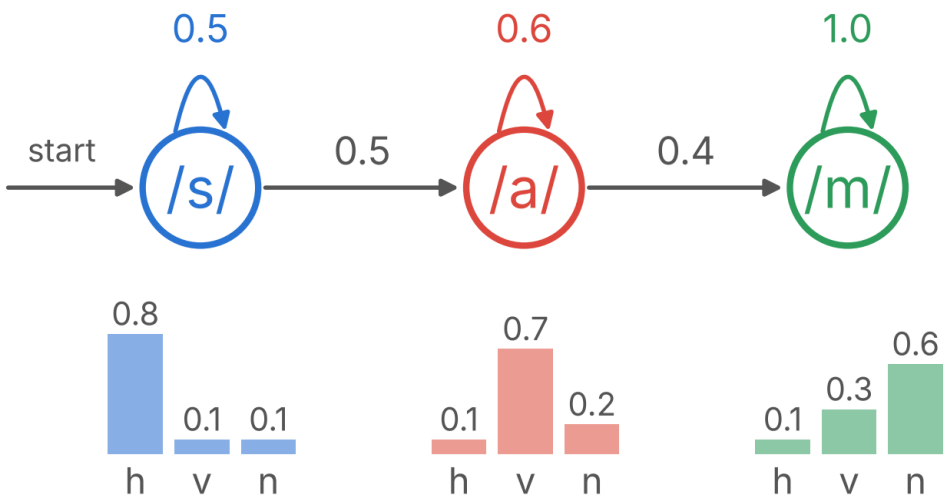
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Backtrace: a ← a ← s ← s, so **s s | a a**: Practice 1, path a. **Check** the sum: the three alignments that end in /a/ give $0.0512 + 0.1024 + 0.0128 = 0.1664$. The best path carries **62%**.

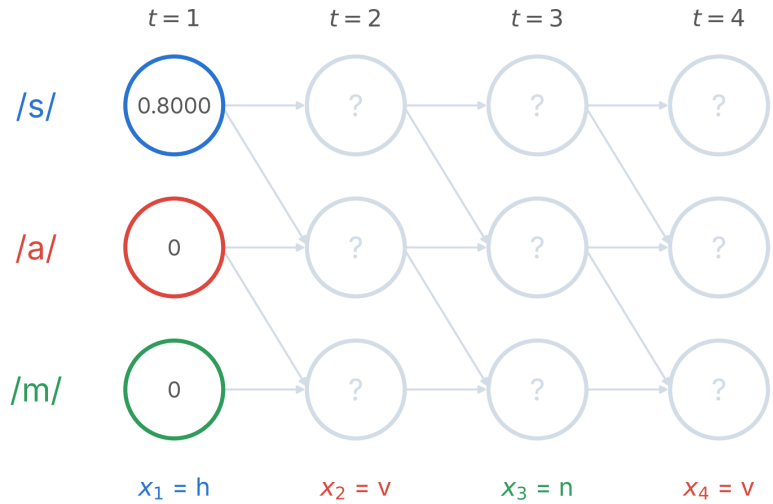
Practice 3: A New Word, Three States (10 min)



A 3-phone word /s a m/ with three observation symbols: h = hiss, v = vowel, n = nasal hum. $\pi = (1, 0, 0)$. We hear $X = (h, v, n, v)$.

1. **Before computing**: which cells must be 0?
2. **Greedy**: pick the best state per frame. Is that a possible path?
3. Fill the **Viterbi** trellis and backtrace. The word must end in /m/.

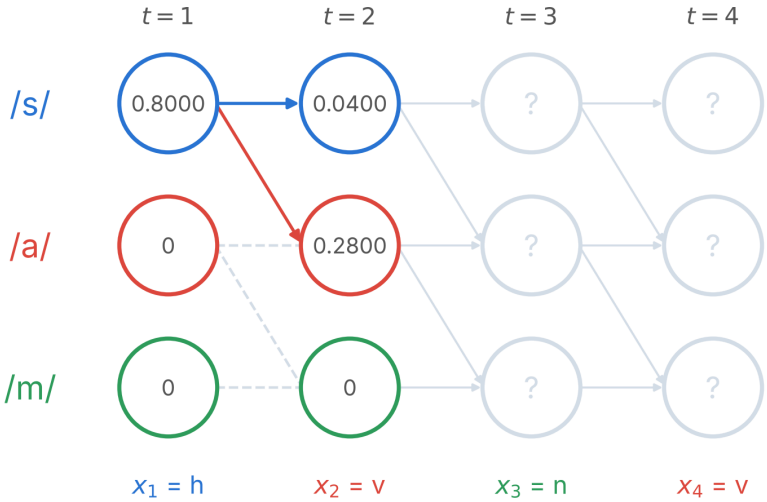
Practice 3: A New Word, Three States (10 min)



1. At $t = 1$ only /s/ is possible (π). At $t = 2$, /m/ is still 0: it can only be entered from /a/, which was 0 at $t = 1$. Every state needs **at least one frame**.

2. Greedy gives **s a m a**: it needs /m/ → /a/, probability 0. **Not a path.**

Practice 3: A New Word, Three States (10 min)



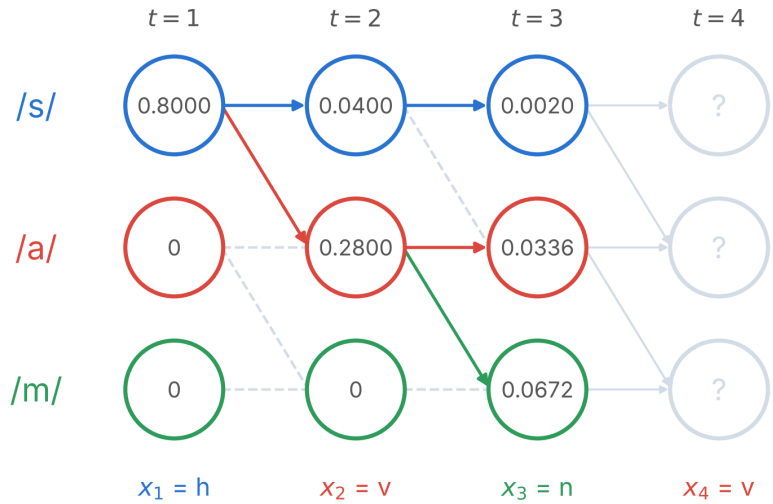
$t = 2$ ($x_2 = v$):

$\delta_2(s) = 0.8 \cdot 0.5 \cdot 0.1 = 0.04$

$\delta_2(a) = 0.8 \cdot 0.5 \cdot 0.7 = 0.28$

$\delta_2(m) = 0$

Practice 3: A New Word, Three States (10 min)



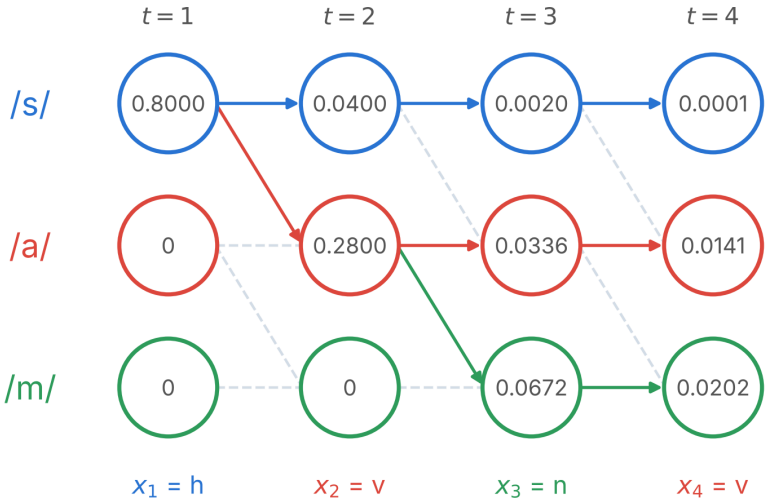
$t = 3$ ($x_3 = n$):

$\delta_3(s) = 0.04 \cdot 0.5 \cdot 0.1 = 0.002$

$\delta_3(a) = \max(0.04 \cdot 0.5,\ 0.28 \cdot 0.6) \cdot 0.2 = 0.0336$, from /a/

$\delta_3(m) = \max(0.28 \cdot 0.4,\ 0 \cdot 1) \cdot 0.6 = 0.0672$, from /a/

Practice 3: A New Word, Three States (10 min)



$t = 4$ ($x_4 = v$, which favours /a/: 0.7 vs 0.3):

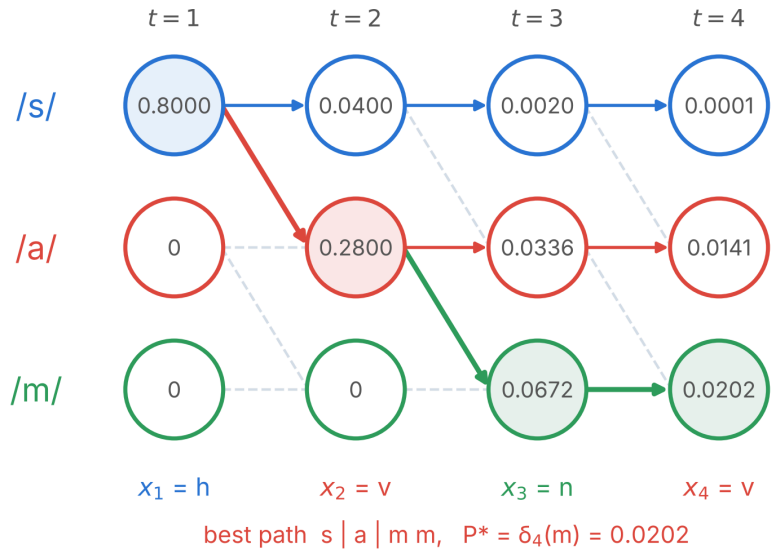
$\delta_4(s) = 0.002 \cdot 0.5 \cdot 0.1 = 0.0001$

$\delta_4(a) = \max(0.002 \cdot 0.5,\ 0.0336 \cdot 0.6) \cdot 0.7 = 0.0141$, from /a/

$\delta_4(m) = \max(0.0336 \cdot 0.4,\ 0.0672 \cdot 1) \cdot 0.3 = 0.0202$, from /m/

Hands up: which cell is the answer, and why not the larger emission?

Practice 3: A New Word, Three States (10 min)



The word must **end in /m/**, so the answer is $\delta_4(m)$, even though /a/ fits the last frame better.
Backtrace: $\psi_4(m) = m$, $\psi_3(m) = a$, $\psi_2(a) = s$.

s | a | m m, $P^* \approx 0.0202$. The last v is explained as a weak emission of /m/: there is no way back to /a/.

Activity 7: Sum or Max? (2 min)

Stand up for Max (Viterbi). **Stay seated for Sum** (forward).

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1. How well does the word *salaam* fit this audio?

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1. How well does the word *salaam* fit this audio? **Sum: the total over all alignments.**

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1. How well does the word *salaam* fit this audio? **Sum: the total over all alignments.**
2. Where does each phone start and end?

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1. How well does the word *salaam* fit this audio? **Sum: the total over all alignments.**
2. Where does each phone start and end? **Max: forced alignment.**

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4. The fastest decoder, in log space, with no logsumexp.

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6. Bonus: a trellis with only one path of non-zero probability.

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6. Bonus: a trellis with only one path of non-zero probability. **Both: sum = max.**

Part 2

GMMs by Hand

Human k-means, Gaussian scores, shares, EM, and training the GMM-HMM

Activity 6: Human k-means (6 min)

Ten volunteers hold the cards **1 2 3 4 5 6 12 13 14 15** and stand under their number on the board. Two more volunteers are the **centres**: blue starts at **1**, red at **4**. Then repeat until nobody switches:

1. **Assign**: every card-holder joins the **nearer** centre.

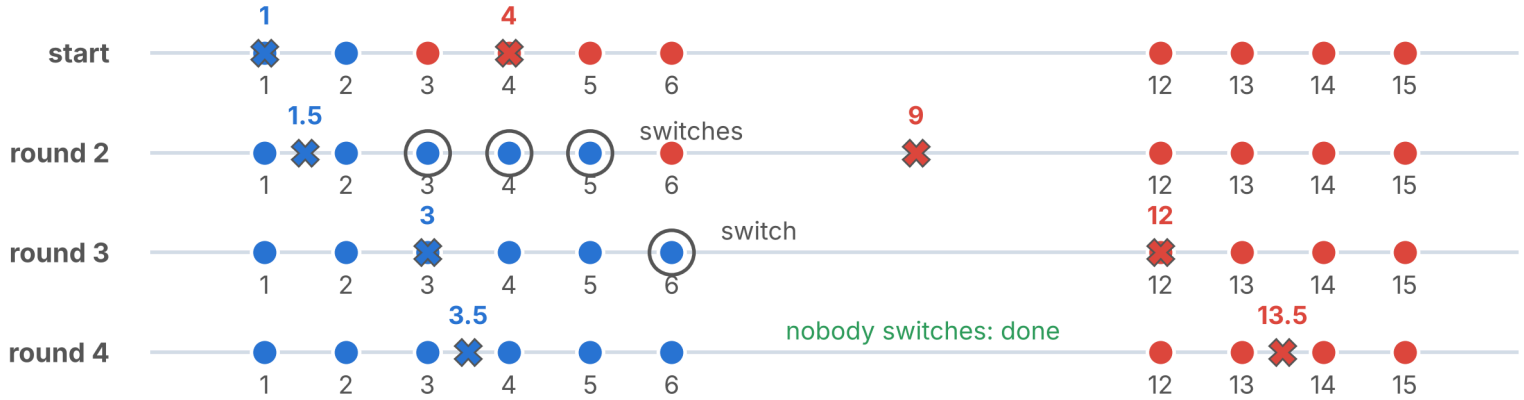
2. **Update**: each group calls out its **average**, and its centre walks there.

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2. **Update**: each group calls out its **average**, and its centre walks there.



Cards 3, 4 and 5 switch in round 2, card **6** switches last, and the centres stop at **3.5** and **13.5**. This is **k-means**: every card belongs 100% to one group. Later today it returns as the hard version of **EM**.

Recap: Scoring a Frame with Gaussians

One bump: measure the distance in σ units, then turn it into a height:

$$z = \frac{x - \mu}{\sigma}, \qquad \mathcal{N}(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-z^2/2} \approx \frac{0.4}{\sigma}\, e^{-z^2/2}$$

A mixture: add the bumps, each times its weight: $p(x) = \sum_m c_m \mathcal{N}(x; \mu_m, \sigma_m)$, with $\sum_m c_m = 1$.

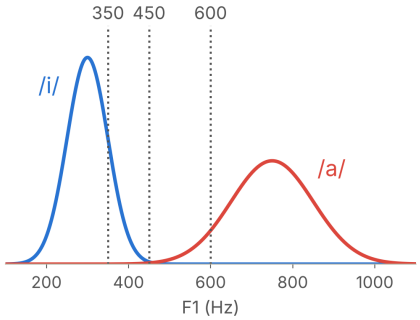
In logs: $\log \mathcal{N} = -\log(\sqrt{2\pi}\,\sigma) - z^2/2$, with $\log \sqrt{2\pi} \approx 0.92$.

z-table, for every practice in Part 2:

z	0	±0.5	±1	±1.5	±2	±2.5	±3	±4
$e^{-z^2/2}$	1	0.882	0.607	0.325	0.135	0.044	0.011	0.00034

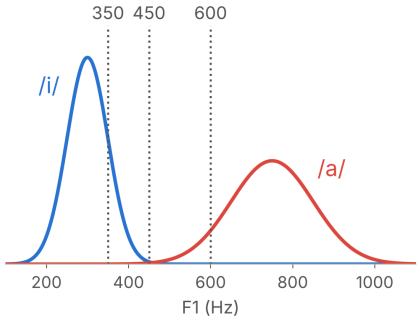
Practice 4: Which Vowel, Frame by Frame? (5 min)

Activity 5 of Lecture 5 again: /i/ has $\mu = 300$, $\sigma = 50$; /a/ has $\mu = 750$, $\sigma = 100$. Score three new frames with the z-table, and pick the winner for each. If z is off the table, take $e^{-z^2/2} \approx 0$.



Practice 4: Which Vowel, Frame by Frame? (5 min)

Activity 5 of Lecture 5 again: /i/ has $\mu = 300$, $\sigma = 50$; /a/ has $\mu = 750$, $\sigma = 100$. Score three new frames with the z-table, and pick the winner for each. If z is off the table, take $e^{-z^2/2} \approx 0$.

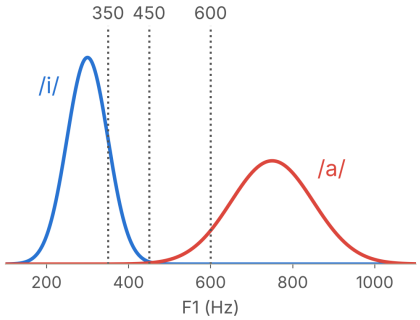


Peak heights $0.4/\sigma$: /i/ $= 0.4/50 = 0.008$, /a/ $= 0.4/100 = 0.004$. **Each cell**: find z, look up $e^{-z^2/2}$ in the z-table, multiply by the peak.

F1	/i/ ($\mu = 300,\ \sigma = 50$)	/a/ ($\mu = 750,\ \sigma = 100$)	wins
350	$z = (350 - 300)/50 = 1$ $0.008 \times 0.607 = 4.9 \times 10^{-3}$	$z = (350 - 750)/100 = -4$ $0.004 \times 0.00034 = 1.4 \times 10^{-6}$	/i/
450	$z = (450 - 300)/50 = 3$ $0.008 \times 0.011 = 8.8 \times 10^{-5}$	$z = (450 - 750)/100 = -3$ $0.004 \times 0.011 = 4.4 \times 10^{-5}$	**/i/, 2×**
600	$z = (600 - 300)/50 = 6$ off the table: ≈ 0	$z = (600 - 750)/100 = -1.5$ $0.004 \times 0.325 = 1.3 \times 10^{-3}$	/a/

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At 450 Hz both are **3σ** away: the narrower /i/ bump is **taller** ($0.4/50 > 0.4/100$), so /i/ wins.

Chicken and Egg: the EM Idea

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- If we knew **which bump each point came from**, we could fit the bumps.

Chicken and Egg: the EM Idea

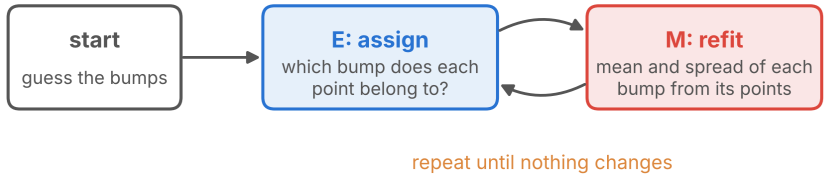
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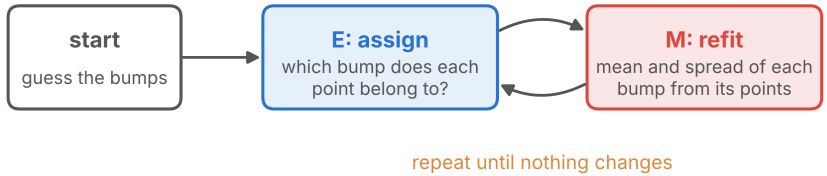
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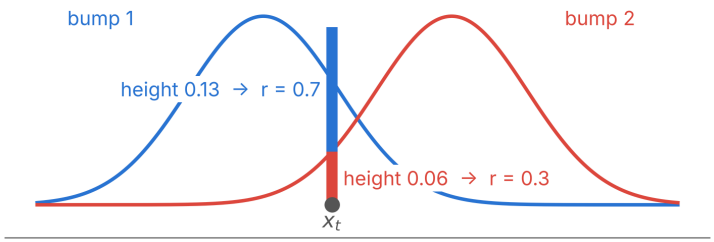


This loop is the **EM** algorithm: **E**xpectation (assign), **M**aximization (refit).

Important
Keep this idea in mind: we will meet the same chicken-and-egg problem again.

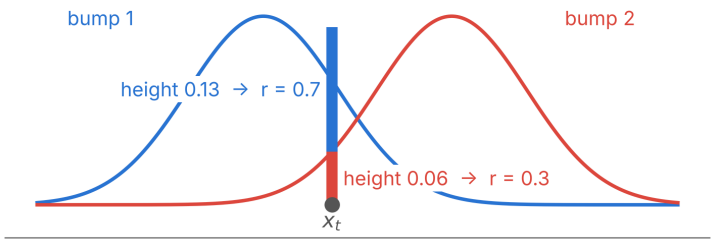
EM, Soft Version: the E-step

In Activity 6, card 6 had to join **one** group. The soft version **shares** each point between the bumps, in proportion to their heights at that point:



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$$r_{tm} = \frac{c_m \, \mathcal{N}(x_t; \mu_m, \sigma_m)}{\sum_k c_k \, \mathcal{N}(x_t; \mu_k, \sigma_k)} \qquad \text{here: } \frac{0.13}{0.13 + 0.06} \approx 0.7$$

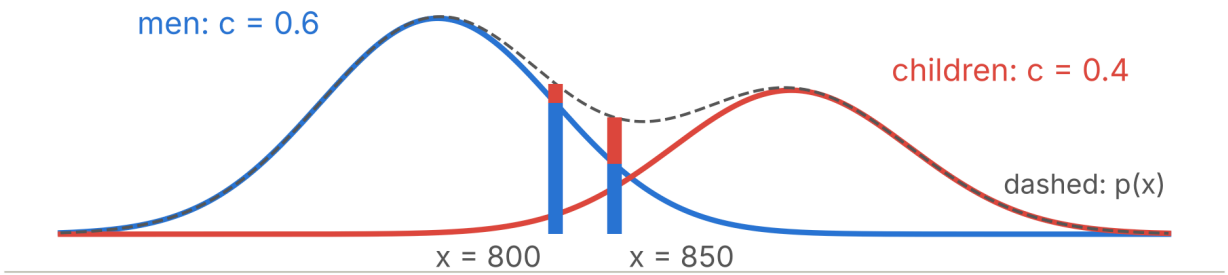
r_{tm} is the **responsibility** of bump m for point x_t; the shares of each point sum to 1. Watch EM run in **Trellis Lab**.

Practice 5: Two Bumps, One Frame (4 min)

The E-step on single frames. A 2-bump mixture for /a/: men $c = 0.6$, $\mu = 700$; children $c = 0.4$, $\mu = 1000$; both $\sigma = 100$. For **x = 800** and **x = 850**, find $p(x)$ and the share of the men's bump.

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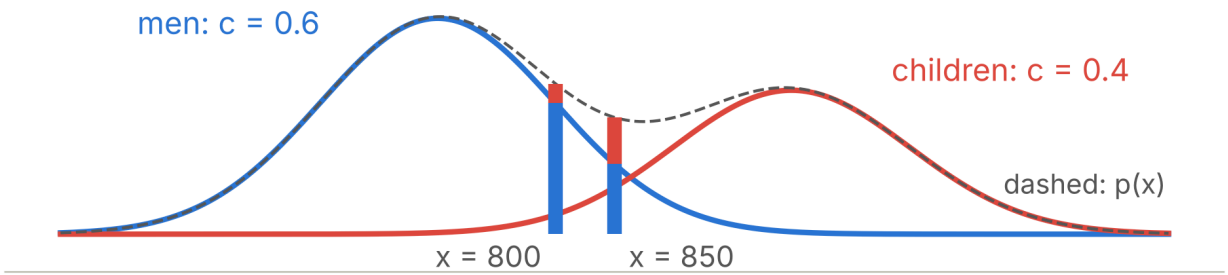
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x	men: $0.6 \cdot 0.004\, e^{-z^2/2}$	children: $0.4 \cdot 0.004\, e^{-z^2/2}$	$p(x)$	r_{men}
800	$z = 1$: 1.46×10^{-3}	$z = -2$: 0.22×10^{-3}	1.67×10^{-3}	**0.87**
850	$z = 1.5$: 0.78×10^{-3}	$z = -1.5$: 0.52×10^{-3}	1.30×10^{-3}	**0.60**

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At 850 the frame is 1.5σ from both bumps, and both bumps have the same σ, so the shares are just the **weights** c_m. The shares of one frame add to 1 over the **bumps**, not over the frames.

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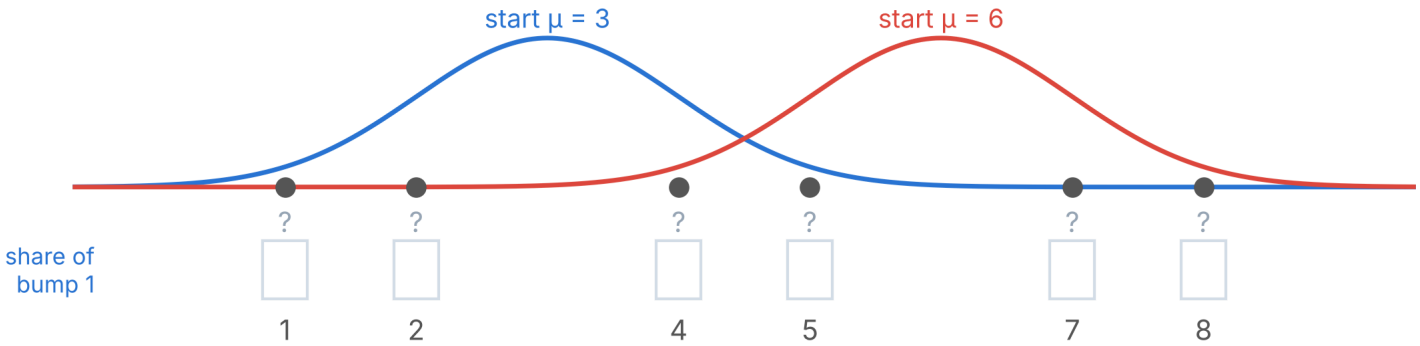
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> **Tip**
> If every share is exactly 0 or 1, the centre update is exactly what the human k-means did in Activity 6.

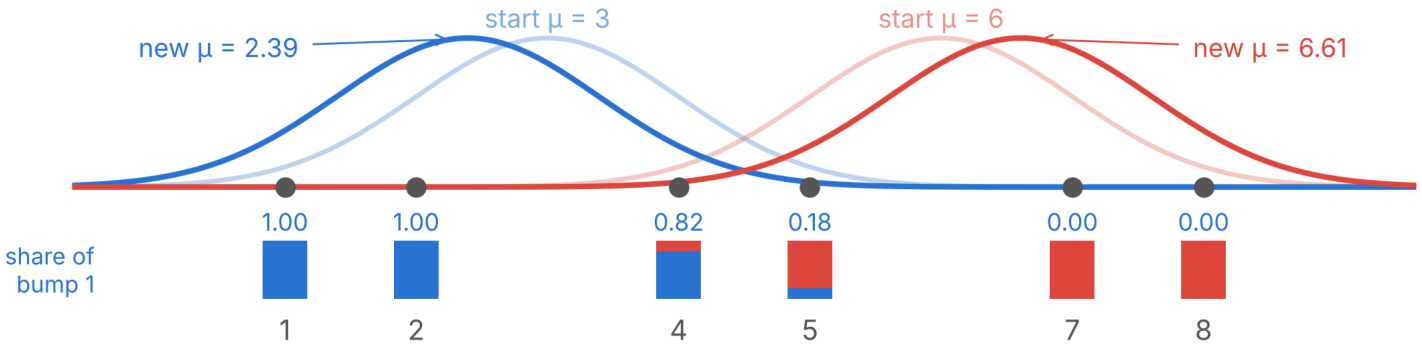
Practice 6: One EM Step by Hand (8 min)



Six points: **1, 2, 4, 5, 7, 8**. Two bumps start at $\mu = (3, 6)$. To keep it short, $\sigma = 1$ and $c = (0.5, 0.5)$ stay **fixed**: we update only the centres.

1. **E-step** for $x = 4$: its share of bump 1. (Equal σ and c cancel, so only $e^{-z^2/2}$ matters.) Then guess the shares of 5, of 1 and 2, and of 7 and 8 without computing.
2. **M-step**: the new μ_1 and μ_2.
3. Compare with **hard** EM (k-means), where 4 and 5 each go 100% to the nearer centre.

Practice 6: One EM Step by Hand (8 min)



1. E-step: 4 is 1σ from bump 1 and 2σ from bump 2, so $r = 0.607/(0.607 + 0.135) = 0.82$. By symmetry 5 gets 0.18; 1 and 2 get ≈ 1; 7 and 8 get ≈ 0.

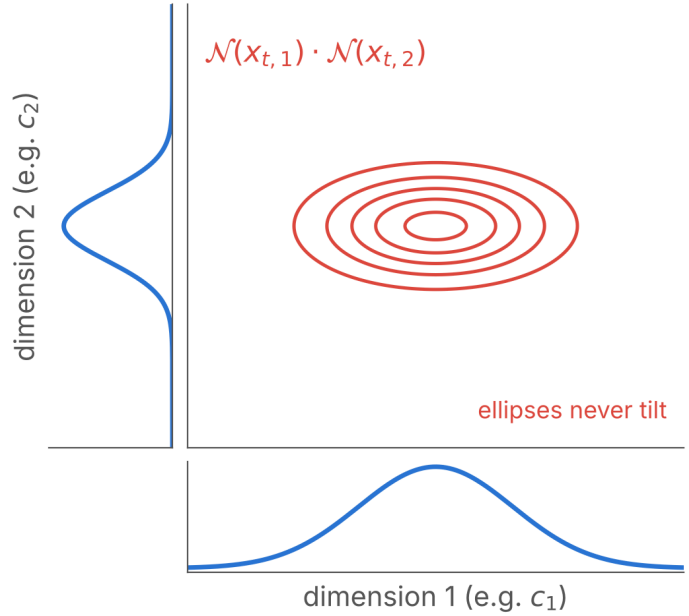
2. M-step: weighted averages, each bump with its own shares:

$$\mu_1 = \frac{1 + 2 + 0.82 \cdot 4 + 0.18 \cdot 5}{1 + 1 + 0.82 + 0.18} = \frac{7.18}{3} = 2.39 \qquad \mu_2 = \frac{0.18 \cdot 4 + 0.82 \cdot 5 + 7 + 8}{3} = \frac{19.82}{3} = 6.61$$

3. Hard EM: 4 → bump 1 and 5 → bump 2, so $\mu = (7/3,\ 20/3) = (2.33,\ 6.67)$. Soft EM moves the centres less, because 4 and 5 are shared.

From 1 Feature to 39

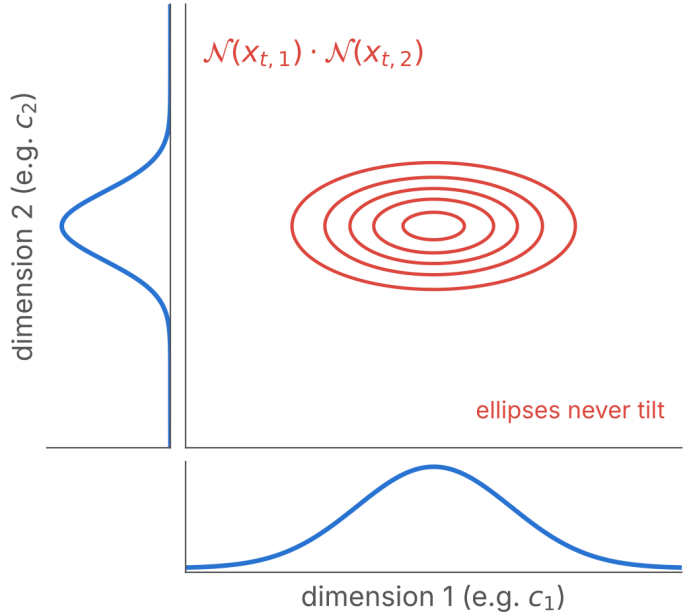
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- Simplest assumption: **inside each bump**, the 39 numbers are **independent**.
- Then each 39-D bump is just a **product** of 39 one-dimensional Gaussians, each with its own μ_d and σ_d:

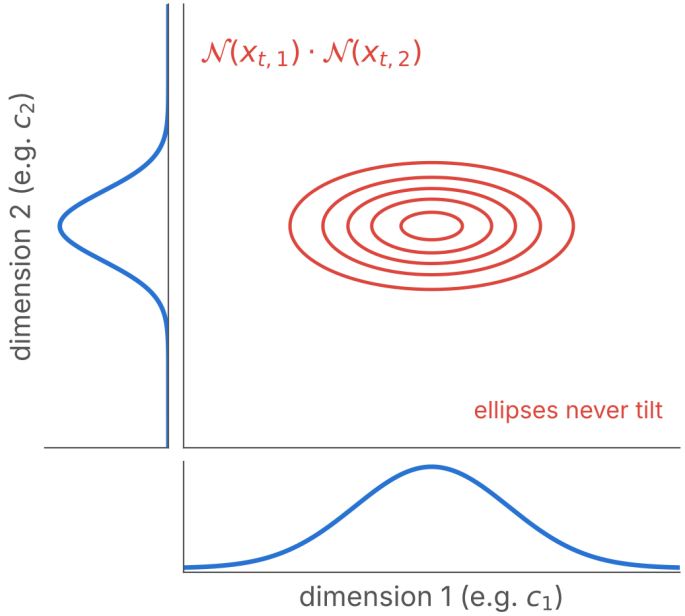


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$$\mathcal{N}(x_t) = \prod_{d=1}^{39} \mathcal{N}(x_{t,d};\ \mu_d, \sigma_d)$$



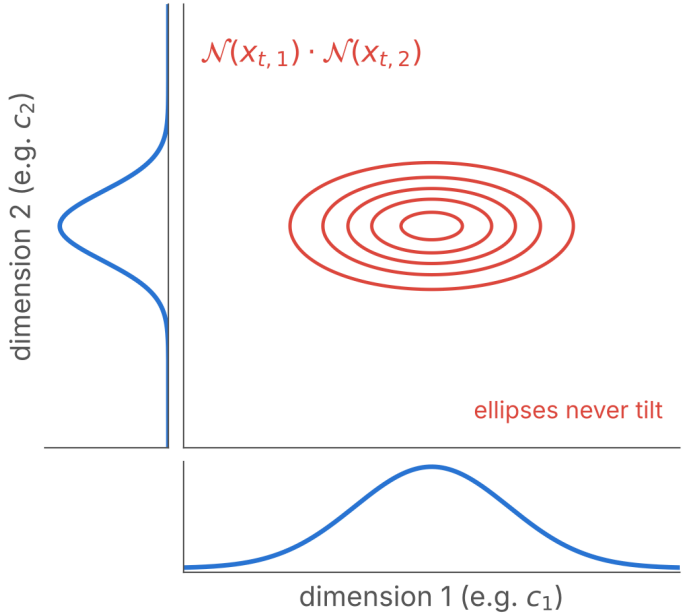
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- Per bump: 39 means and 39 variances, a **diagonal covariance**. A mixture of many such bumps can still follow correlated data.

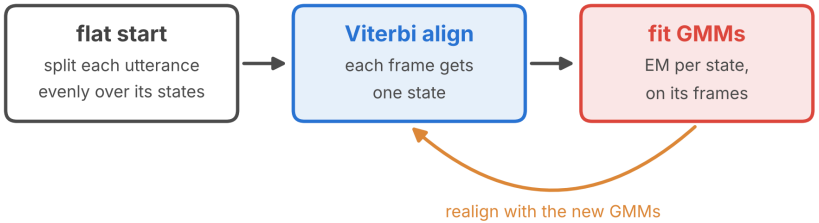


Back into the HMM

Each HMM state j gets its **own** mixture:

$$b_j(x_t) = \sum_{m=1}^{M} c_{jm}\,\mathcal{N}(x_t;\mu_{jm},\sigma_{jm})$$

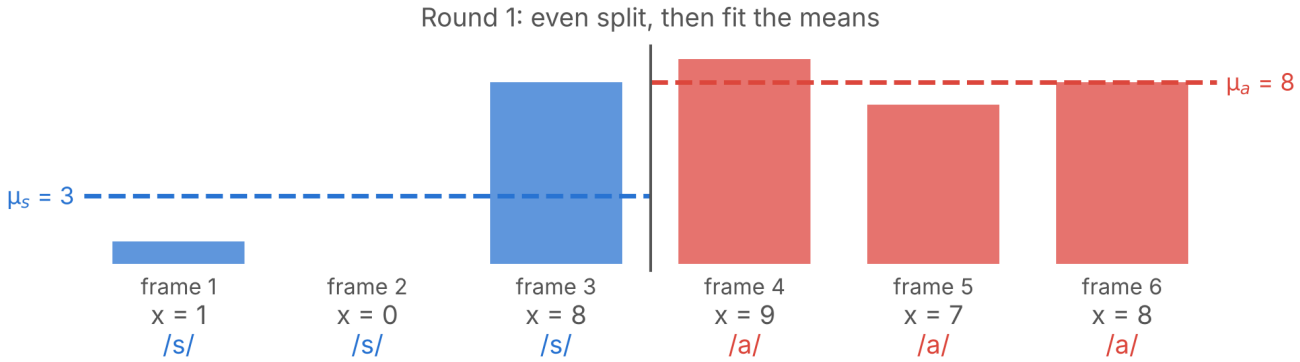
The same chicken-and-egg, one level up: we don't know which **state** produced each frame. So we start from an even split, then alternate:



repeat until the alignment stops changing, then split mixtures (M → 2M) and repeat

> **Note**
> This is **Viterbi training** (hard). The soft version, with shares from the forward pass plus a backward pass, is **Baum–Welch**: Rabiner (1989), Problem 3.

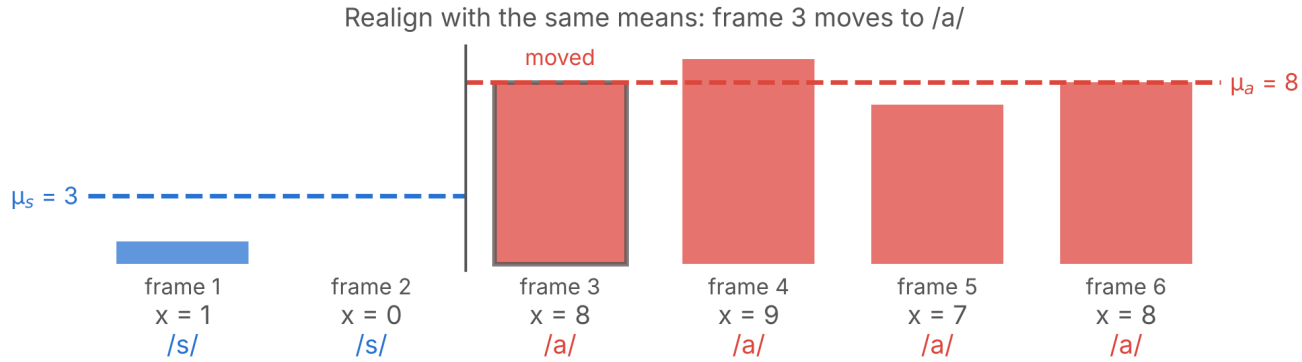
Practice 8: Train a GMM-HMM by Hand (10 min)



Six frames of /s a/ with one made-up feature, **voicing** (0 = none, 9 = strong). One Gaussian per state, same σ for both, so a frame simply goes to the state whose **mean is nearer**, just like human k-means.

Step 1, flat start: split evenly, 3 frames each. Compute the two means. Then **step 2**: which frame should change state?

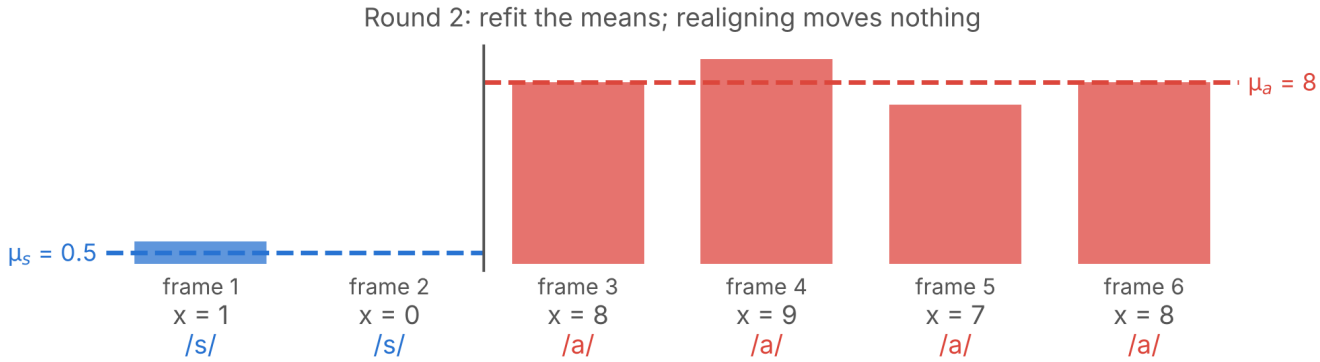
Practice 8: Train a GMM-HMM by Hand (10 min)



Means: $\mu_s = (1+0+8)/3 = 3,\ \ \mu_a = (9+7+8)/3 = 8.$

Realign: frame 3 has $x = 8$. Its distance to μ_s is 5, to μ_a is 0, so it **moves to /a/**. Every other frame is already nearer its own mean.

Practice 8: Train a GMM-HMM by Hand (10 min)



Refit: $\mu_s = (1+0)/2 = 0.5,\ \ \mu_a = (8+9+7+8)/4 = 8$. **Realign**: nothing moves, so we stop.

- That is the whole loop: **align → refit the means → align again**, until nothing moves.
- One wrong frame in the even split pulled μ_s up to 3; one round fixed it.

Part 3

Putting It Together

The GMM-HMM and its limits

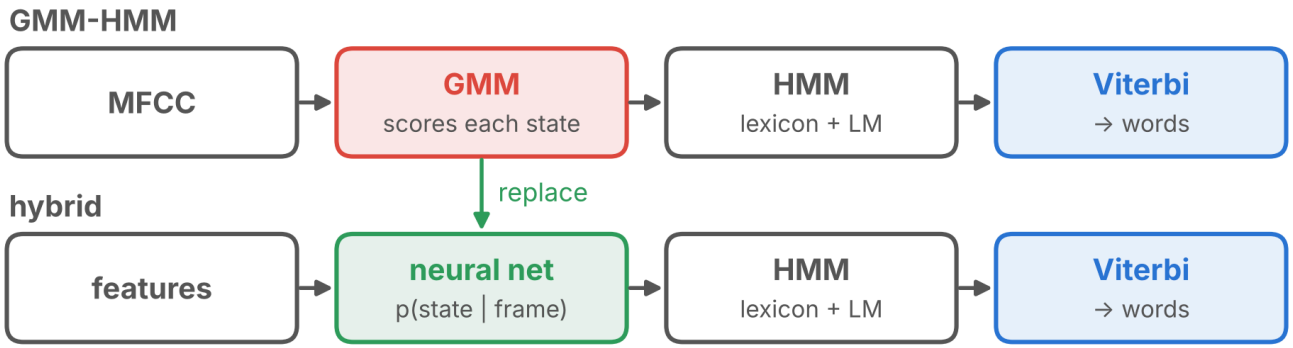
Why the GMM Is the Weak Link

The GMM ...	A neural network ...
scores **one frame** at a time	reads a **window of frames** around it
needs **MFCCs**, hand-made features, so that diagonal Gaussians fit	**learns its own features** from log-mel
fits each state on **its own frames only**, so it never learns what tells /s/ from /a/	is trained to **tell the states apart**

The HMM and Viterbi can stay: the first step replaces only the scoring (next slide).

The Bridge to Neural ASR: the Hybrid DNN-HMM

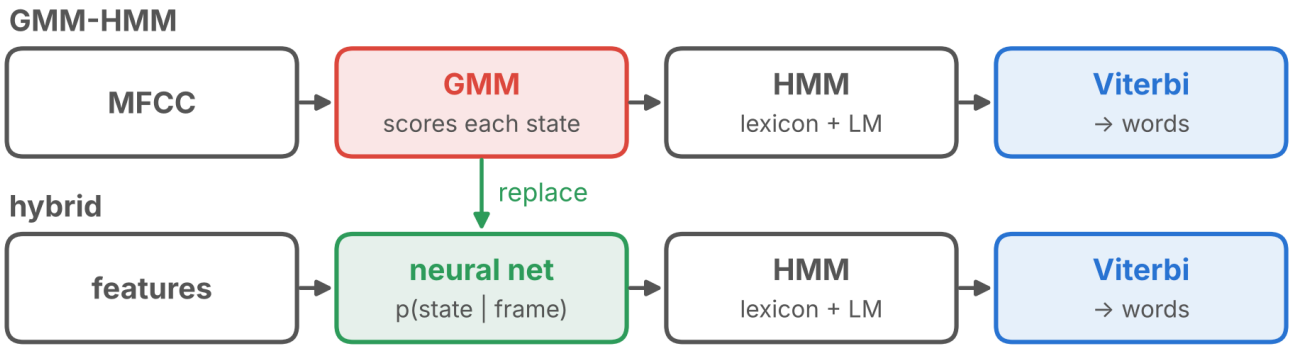
Keep the HMM and Viterbi. Replace **only the GMM** with a neural network that reads the frame and outputs a probability for each state.



the catch: training the net needs a state label for every frame, and those labels come from a GMM-HMM forced alignment

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the catch: training the net needs a state label for every frame,
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Important
Can we train on (audio, transcript) pairs directly, and never fix an alignment? That is **CTC**: next lecture.

Practice 9: Exam-Style Problem (10 min)

A 2-state word /s/ → /a/: $\pi = (1, 0)$, $a_{ss} = 0.5$, $a_{sa} = 0.5$, $a_{aa} = 1$. Each state emits **one Gaussian** with $\sigma = 1$: /s/ has $\mu = 0$, /a/ has $\mu = 4$. We observe $X = (0, 2, 4)$.

a. All six $b_j(x_t)$, with the z-table and $1/\sqrt{2\pi} \approx 0.4$. **b.** $P(X \mid \lambda)$ with forward (the word ends in /a/). **c.** The best path with Viterbi. **d.** Without any table: why is $b_s(2) = b_a(2)$, and what decides frame 2?

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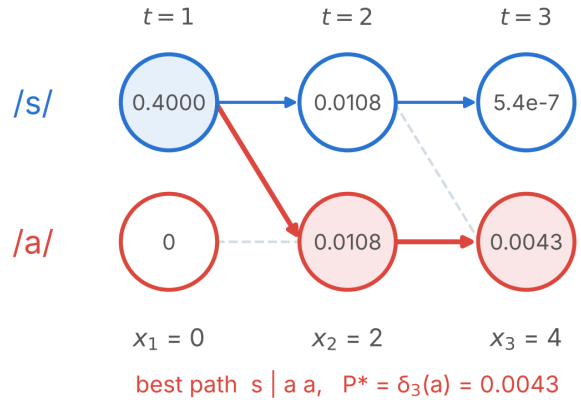
a. $b_s = (0.4,\ 0.054,\ 0.0001)$ and $b_a = (0.0001,\ 0.054,\ 0.4)$: $z = 0$ gives 0.4, $z = \pm 2$ gives $0.4 \cdot 0.135$, $z = \pm 4$ is almost 0.

b. $\alpha_1 = (0.4,\ 0)$; $\alpha_2(s) = 0.4 \cdot 0.5 \cdot 0.054 = 0.0108$, $\alpha_2(a) = 0.4 \cdot 0.5 \cdot 0.054 = 0.0108$;
$\alpha_3(a) = (0.0108 \cdot 0.5 + 0.0108 \cdot 1) \cdot 0.4 = \mathbf{0.0065}$

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c. $\delta_3(a) = \max(0.0108 \cdot 0.5,\ 0.0108 \cdot 1) \cdot 0.4 = 0.0043$, from /a/: **s | a a**.

d. Frame 2 is 2σ from both centres, and the σ's are equal, so the emissions tie. The **transitions** break the tie: from /a/ the next step is free ($a_{aa} = 1$), from /s/ it still costs $a_{sa} = 0.5$. The path s s | a is half as likely (0.0022).

Practice 10: True or False? (5 min)

Stand up for True. Stay seated for False.

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1. A Gaussian density can be larger than 1.

Practice 10: True or False? (5 min)

Stand up for True. Stay seated for False.

1. A Gaussian density can be larger than 1. **True: the peak is $0.4/\sigma$, so $\sigma = 0.1$ gives 4. It is a density, not a probability.**

Practice 10: True or False? (5 min)

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7. Viterbi training and Baum–Welch differ only in hard vs. soft frame shares. **True: best path vs. all paths, the same max vs. sum as in Part 1.**

What to Take Away

- **Viterbi** is the forward algorithm with max instead of sum, plus backpointers. It finds the best alignment in $O(N^2T)$, needs no logsumexp, and with a known transcript gives **forced alignment**.
- A **GMM** scores a continuous frame: Gaussians measure distance in σ units, mixtures handle several groups, and **EM** fits them by alternating assignment and refitting. With a diagonal covariance, each 39-D bump is a product of 39 one-dimensional Gaussians.
- A **GMM-HMM** is trained by alternating Viterbi alignment and GMM fitting. Its assumptions motivate the neural models in the rest of the course.

Next lecture: **CTC**, training on (audio, transcript) pairs without a fixed alignment.

> **Important**
> Reading: Rabiner (1989), Problems 2 and 3 and the section on continuous mixture densities.

Learn More

Course Homepage · Rabiner (1989): A Tutorial on Hidden Markov Models · J&M: Appendix A, Hidden Markov Models · scikit-learn: Gaussian mixture models · Montreal Forced Aligner · hmmlearn